MTP: The New Way to Exchange Data

This article explores the differences between the older Mass Storage Class (MSC) and the newer Media Transfer Protocol (MTP) methods of sharing embedded-device memory when connected to a PC.

Storage is a key component of any embedded handheld device like a mobile phone or other convergence devices. To transfer the data stored in these devices, technologies such as USB or Bluetooth are used, the former being more predominant. Generally, when the USB shares the memory of a mobile device, it is shared as a drive on the host PC. The USB class used is known as Mass Storage Class (MSC), and it allows you to manage the drive completely like a local hard drive connected to the PC. But in recent Android versions (Ice Cream Sandwich and later), a newer class named MTP (Media Transfer Protocol) is made the default to share mobile memory over USB to a host PC. So what necessitated this change? Let's take a brief look at the two USB classes to understand why MTP is preferred over MSC.

Before going into the advantages of MTP, let us first explore how these two protocols work, with the help of Linux.

An overview of USB MSC

The MSC protocol defines how data exchange can take place between a USB host and a USB-compliant device. Whenever you use the ubiquitous pen drive for all your data-transfer activities, MSC is the underlying protocol. USB devices using the MSC class are equivalent to an external hard drive connected to a host. These devices can be used just like in-PC drives, to drag and drop files. Thus, the USB host can interact with different flash/hard drives without having any knowledge about the underlying storage system.

Linux USB MSC architecture

Figure 1 shows the block-level architecture when an MSC device is connected to a standard Linux host. The Linux device's MSC architecture consists of two parts: the storage subsystem and the USB subsystem. In the Linux framework, the Virtual File System (VFS) layer is used to abstract the storage and the USB layers. The gadget driver, being USB controller-specific, collects the USB transfers and identifies to which function the transfer belongs (e.g., the Ethernet packet, the MSC packet, etc). If it is an MSC packet, the gadget driver sends the packet to the mass storage driver. Inside the class driver, the SCSI commands are decoded, and appropriate storage operations are done using the VFS layer. SCSI (Small Computer Systems Interface) is a standard for the transfer of data between two devices. The entire blocks are implemented in kernel space.

On the host side, the USB core drivers detect the USB device and expose it as a block device. The data is transferred to the block layer as SCSI commands, with the help of the storage driver.

An overview of the Media Transfer Protocol (MTP)

MTP was introduced by Microsoft to enable data exchange between a USB host (initiators) and devices (responders) with 'intelligent' storage capabilities (those with some sort of user interactivity). These include smartphones, tablets and portable audio players. 'Media'